Assignment 2

STAT-540

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1 Shreve Exercise 1.10

Let $\mathbb{P}$ be the uniform (Lebesgue) measure on $\Omega = [0, 1]$. Define

$$Z(\omega) = \begin{cases} 0 \text{ if } 0 \leq \omega < \frac{1}{2}, \\ 2 \text{ if } \frac{1}{2} \leq \omega \leq 1. \end{cases}$$

For $A \in \mathcal{B}[0, 1]$, define

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega).$$

(i) Show that $\widetilde{\mathbb{P}}$ is a probability measure.

(ii) Show that if $\mathbb{P}(A) = 0$, then $\widetilde{\mathbb{P}}(A) = 0$. We say that $\widetilde{\mathbb{P}}$ is *absolutely continuous* with respect to $\mathbb{P}$.

(iii) Show that there is a set A for which $\widetilde{\mathbb{P}}(A) = 0$, but $\mathbb{P}(A) > 0$. In other words, $\widetilde{\mathbb{P}}$ and $\mathbb{P}$ are not equivalent.

Solution:

1. To prove that P_tilde is a probability measure, we need to verify the following properties:
   a. Non Negativity:

$$Since\ \mathbb{z}(w) \geq 0, for\ all\ w\ and\ \mathrm{P}\ is\ a\ measure, we\ have:$$

$$\tilde{P}(A) = \int_A Z(w) dP(w) \geq 0$$

   b. Normalization:

$$\tilde{P}([0,1]) = \int_0^1 Z(w) dP(w) = \int_0^{\frac{1}{2}} 0 * dP(w) + \int_{\frac{1}{2}}^1 2 * dP(w) = 0 + 2\mathrm{P}\left(\left[\frac{1}{2,1}\right]\right)$$

$$= 2 * \frac{1}{2} = 1$$

c. Countable additivity:

$$P\left(\bigcup_n A_n\right) = \int \left(\bigcup_n A_n\right) Z(w)dP(w) = \sum_n \int (A_n)Z(w)dP(w)$$

$$by\ countable\ additivity\ of\ Lebesgue\ Integral$$

$$= \sum_n \tilde{P}(A_n)$$

2. If P(A) = 0, then A has a Lebesgue measure zero

$$\tilde{P}(A) = \int_A Z(w)dP(w) = \sum_n \int (A_n)Z(w)dP(w)$$

$$since\ Z(w)\ is\ bounded\ by\ (0 \le Z(w) \le 2)\ and\ P(A) = 0, we\ have:$$

$$\tilde{P}(A) \le \int_A 2dP(w) = 2 * P(A) = 2 * 0 = 0$$

$$since\ \tilde{P}(A) \ge 0, by\ non - negativity, we\ conclude: \tilde{P}(A) = 0$$

3. To find the posible A that follow that condition, consider where A = [0,0.5):

$$Calculate\ \tilde{P}(A): \tilde{P}\left(\left[0, \frac{1}{2}\right]\right) = \int_0^{\frac{1}{2}} Z(w)dP(w) = \int_0^{\frac{1}{2}} 0 * dP(w) = 0$$

$$Calculate\ P(A): P\left(\left[0, \frac{1}{2}\right]\right) = \frac{1}{2} > 0$$

$$Therefore, A = \left[0, \frac{1}{2}\right]\ satisfies\ \tilde{P}(A) = 0, and\ P(A) = \frac{1}{2}, which\ is > 0$$

2 Shreve Exercise 1.14

(Change of measure for an exponential random variable) Let X be a nonnegative random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with the *exponential distribution*, which is

$$\mathbb{P}\{X \le a\} = 1 - e^{-\lambda a}, a \ge 0,$$

where λ is a positive constant. Let $\tilde{\lambda}$ be another positive constant, and define

$$Z = \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X}$$

. Define $\widetilde{\mathbb{P}}$ by

$$\widetilde{\mathbb{P}}(A) = \int_A Z d\mathbb{P} \text{ for all } A \in \mathcal{F}.$$

(i) Show that $\widetilde{\mathbb{P}}(\Omega) = 1$.

(ii) Compute the cumulative distribution function

$$\widetilde{\mathbb{P}}\{X \le a\} \text{ for } a \ge 0$$

for the random vatiable X under the probability measure $\widetilde{\mathbb{P}}$.

Solution:

1. To show that P_tilde over the probability space is equal to 1:

$$\tilde{P}(\Omega) = \int_\Omega Z dP = \int_\Omega \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X} dP = \int_0^\infty \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X} \lambda e^{-(\lambda)X} dx =$$

$$\int_0^\infty \tilde{\lambda} e^{-(\tilde{\lambda})X} dx = \tilde{\lambda}\left(\frac{-1}{\tilde{\lambda}}\right)\left(e^{-\tilde{\lambda}\,\infty} - e^{-\tilde{\lambda}\,0}\right) = 1$$

2. Cumulative distribution for X<=a, for a > 0:

$$\tilde{P}\{X \le a\} = \int_0^a \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X} dP = \int_0^a \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X} \lambda e^{-(\lambda)X} dx = \tilde{\lambda}\left(\frac{-1}{\tilde{\lambda}}\right)\left(e^{-\tilde{\lambda}\,a} - e^{-\tilde{\lambda}\,0}\right)$$
$$= 1 - e^{-\tilde{\lambda}\,a}$$

3 Shreve Exercise 2.2

Independence of random variables can be affected by changes of measure. To illustrate this point, consider the space of two coin tosses $\Omega_2 = \{HH, HT, TH, TT\}$, and let stock prices be given by

$$S_0 = 4,\ S_1(H) = 8,\ S_1(T) = 2,$$
$$S_2(HH) = 16,\ S_2(HT) = S_2(TH) = 4,\ S_2(TT) = 1.$$

Consider two probability measures given by

$$\widetilde{\mathbb{P}}(HH) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(HT) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(TH) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(TT) = \tfrac{1}{4},$$
$$\mathbb{P}(HH) = \tfrac{4}{9},\ \mathbb{P}(HT) = \tfrac{2}{9},\ \mathbb{P}(TH) = \tfrac{2}{9},\ \mathbb{P}(TT) = \tfrac{1}{9}.$$

Define the random variable

$$X = \begin{cases} 1 \text{ if } S_2 = 4, \\ 0 \text{ if } S_2 \neq 4. \end{cases}$$

(i) List all the sets in $\sigma(X)$.

(ii) List all the sets in $\sigma(S_1)$.

(iii) Show that $\sigma(X)$ and $\sigma(S_1)$ are independent under the probability measure $\widetilde{\mathbb{P}}$.

(iv) Show that $\sigma(X)$ and $\sigma(S_1)$ are not independent under the probability measure $\mathbb{P}$.

(v) Under $\mathbb{P}$, we have $\mathbb{P}\{S_1 = 8\} = \frac{2}{3}$ and $\mathbb{P}\{S_1 = 2\} = \frac{1}{3}$. Explain intuitively why, if you are told that $X = 1$, you would want to revise your estimate of the distribution of S_1.

Solution:

1. List all the sets in sigma(X):

The σ − algebra generated by X contains all sets that can be expressed in terms of X.

Since X takes values {0,1}:

− $\{X = 1\}$: $\{HT, TH\}$ $(where\ S_2 = 4)$

− $\{X = 0\}$: $\{HH, TT\}$ $(where\ S_2 \neq 4)$

$$\sigma(X) = \{\emptyset, \{HT, TH\}, \{HH, TT\}, \Omega_2\}$$

2. List all the sets in sigma(S1):

The σ
− algebra generated by S_1 contains all sets that can be expressed in terms of S_1.

Since X takes values {2,8}:

− $\{S_1 = 2\}$: $\{TH, TT\}$ $(first\ toss\ is\ tail)$

$$- \{S_1 = 8\}: \{HT, HH\} \ (first\ toss\ is\ head)$$

$$\sigma(X) = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega_2\}$$

3. Show that sigma x and sigma s1 are independent under P_tilde:

   $The\ two\ \sigma - algebras\ are\ independent\ if\ for\ every\ A \in \sigma(X)\ and\ B \in \sigma(S_1)$:

   $$\tilde{P}(A \cap B) = \tilde{P}(A) * \tilde{P}(B)$$

   Lets check all non-trivial combinations:

   - Case 1:

   $$A = \{HT, TH\}, B = \{HH, HT\}$$

   $$\tilde{P}(A \cap B) = \tilde{P}(HT) = \frac{1}{4}$$

   $$\tilde{P}(A) = \tilde{P}(HT) + \tilde{P}(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

   $$\tilde{P}(B) = \tilde{P}(HH) + \tilde{P}(HT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

   $$\tilde{P}(A) * \tilde{P}(B) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

   - Case 2:

   $$A = \{HT, TH\}, B = \{TH, TT\}$$

   $$\tilde{P}(A \cap B) = \tilde{P}(TH) = \frac{1}{4}$$

   $$\tilde{P}(A) = \tilde{P}(HT) + \tilde{P}(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

   $$\tilde{P}(B) = \tilde{P}(TH) + \tilde{P}(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\tilde{P}(A) * \tilde{P}(B) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

- Case 3:

$$A = \{HH, TT\}, B = \{HH, HT\}$$

$$\tilde{P}(A \cap B) = \tilde{P}(HH) = \frac{1}{4}$$

$$\tilde{P}(A) = \tilde{P}(HH) + \tilde{P}(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\tilde{P}(B) = \tilde{P}(HH) + \tilde{P}(HT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\tilde{P}(A) * \tilde{P}(B) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

- Case 4:

$$A = \{HH, TT\}, B = \{TH, TT\}$$

$$\tilde{P}(A \cap B) = \tilde{P}(TT) = \frac{1}{4}$$

$$\tilde{P}(A) = \tilde{P}(HH) + \tilde{P}(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\tilde{P}(B) = \tilde{P}(TH) + \tilde{P}(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\tilde{P}(A) * \tilde{P}(B) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

We prove that both sigmas are independent.

4. Show that sigma x and sigma s1 are not independent under P:

$The\ two\ \sigma - algebras\ are\ independent\ if\ for\ every\ A \in \sigma(X)\ and\ B \in \sigma(S_1)$:

$$\tilde{P}(A \cap B) = \tilde{P}(A) * \tilde{P}(B)$$

Lets check all non-trivial combinations:

- Case 1:

$$A = \{HT, TH\}, B = \{HH, HT\}$$

$$P(A \cap B) = P(HT) = \frac{2}{9}$$

$$P(A) = P(HT) + P(TH) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$$

$$P(B) = P(HH) + P(HT) = \frac{4}{9} + \frac{2}{9} = \frac{6}{9}$$

$$P(A) * P(B) = \frac{4}{9} * \frac{6}{9} = \frac{24}{81}$$

$$Since: P(A \cap B) \neq P(A) * P(B), both\ sigmas\ are\ not\ independent\ under\ P$$

5. Explain intuitively why, if you are told that X = 1, you would want to revise your estimate of the distribution of $S1$:

$$P\{S_1 = 8\} = P(HH) + P(HT) = \frac{6}{9}$$

$$P\{S_1 = 2\} = P(TH) + P(TT) = \frac{3}{9}$$

Given X=1, which means S2 = 4, so the outcome is either: HT or TH:

$$P\{S_1 = 8, X = 1\} = P(HT) = \frac{2}{9}$$

$$P\{X = 1\} = P(HT) + P(TH) = \frac{4}{9}$$

$$So\ P(S_1 = 2 \mid X = 1) = \frac{P(S1 = 8, X = 1)}{P(X = 1)} = \frac{1}{2}$$

So from having probabilities of P(S1=8) =6/9 - P(S1=2) =3/9, after having S2 = 4, then it both changes to ½ for both. Under P the outcomes HH, HT, TH, TT are not equally likely (they have probabilities 4/9, 2/9, 2/9, 1/9). Learning that S_2 = 4 rules out HH and TT, leaving only HT and TH, which under $\mathbb{P}$ have equal probability (both 2/9). This changes your belief about S_1 from favoring S_1 = 8 to being neutral between S_1 = 8 and S_1 = 2.

This demonstrates that dependence under $\mathbb{P}$ means information about X provides information about S_1.

4 Shreve Exercise 4.8

(Solving the Vasicek equation) The Vasicek interest rate stochastic equation is

$$dR(t) = (\alpha - \beta R(t))dt + \sigma dW(t), \tag{1}$$

where α, β, and σ are positive constants. The solution to this equation is given in Example 4.4.10. This exercise shows how to derive this solution.

(i) Use (1) and the Itô-Doeblin formula to compute $d\left(e^{\beta t}R(t)\right)$. Simplify it so that you have a formula for $d\left(e^{\beta t}R(t)\right)$ that does not involve $R(t)$.

(ii) Integrate the equation you obtained in (i) and solve for $R(t)$ to obtain

$$R(t) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right) + \sigma e^{-\beta t}\int_0^t e^{\beta s}dW(s).$$

Solution:

1. Use Itô-Doeblin formula to compute d(e^(βt)R(t))

$$Y(t) = e^{\beta t} * R(t)$$

Itô's formula: For $Y(t) = f(t, R(t))$, we have: $d(Y) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial R}dR + \frac{1}{2}\frac{\partial^2 f}{\partial^2 R^2}(dR)^2$

$$\text{For } f(t, R) = e^{\beta t} * R,$$
$$\frac{\partial f}{\partial t} = \beta e^{\beta t} * R,$$
$$\frac{\partial f}{\partial R} = e^{\beta t},$$
$$\frac{\partial^2 f}{\partial^2 R^2} = 0$$

$$From\ equation\ (1): (dR)^2 = \sigma^2(dW)^2 = \sigma^2 dt\ (using\ the\ rule\ (dW)^2 = dt$$

$$d\left(\, e^{\beta t} * \mathrm{R(t)}\right) = (\beta e^{\beta t} * \mathrm{R(t)}dt) + e^{\beta t}dR(t) + \frac{1}{2}(0)dt$$
$$d\left(\, e^{\beta t} * \mathrm{R(t)}\right) = (\beta e^{\beta t} * \mathrm{R(t)}dt) + e^{\beta t}\left[\left(\alpha - \beta R(t)\right)dt + \sigma dW(t)\right]$$

$$d\left(\, e^{\beta t} * \mathrm{R(t)}\right) = (\beta e^{\beta t} * \mathrm{R(t)}dt) + \alpha e^{\beta t}dt - (\beta e^{\beta t} * \mathrm{R(t)}dt) \; + \sigma e^{\beta t}dW(t)$$
$$= \; \alpha e^{\beta t}dt \; + \sigma e^{\beta t}dW(s)$$

2. Integrate and solve for R(t)

$$\int_0^t d\left(\, e^{\beta t} * \mathrm{R(s)}\right) \; = \int_0^t \alpha e^{\beta s}dt \; + \sigma e^{\beta s}dW(s)$$

$$e^{\beta t}R(t) - e^0 R(0) = \frac{\alpha}{\beta}\, e^{\beta t} - \frac{\alpha}{\beta} \; + \int_0^t \sigma e^{\beta s}dW(s)$$

$$e^{\beta t}R(t) = \; R(0) + \frac{\alpha}{\beta}\left(\, e^{\beta t} - 1\right) \; + \int_0^t \sigma e^{\beta s}dW(s)$$

$$multiply\ by\ e^{-\beta t}$$

$$R(t) = \; R(0)e^{-\beta t} + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right) \; + \sigma e^{-\beta t}\int_0^t e^{\beta s}dW(s)$$